CSE340: Computer Architecture
Spring 2026
Quiz - 01
Duration: 25 minutes

BRAC UNIVERSITY
Inspiring Excellence

A

Name: Solution	ID:	Section: 23

1. Write T (True) or F (False) beside the questions. 1 mark each. **[1 x 3 = 3]**

a. Using multicore processors cannot reduce execution time below the serial portion of the program. T

b. "Make the common case fast" suggests optimizing the rarest operations for maximum impact F

c. Using arithmetic mean to average performance ratios can bias results toward benchmarks with large ratios. T

2. An AMD CPU runs a Discord program with a clock rate of 2.5GHZ that has the following instruction mix:

Instruction Class	Instruction Count	CPI
Load	$1x10^6$	2
Multiplication	$2x10^6$	3
Branch	$4x10^6$	1

a. Find the Average CPI for this program. **[3]**
b. Find the CPU execution Time in milliseconds. **[2]**
c. If the CPI of multiplication instruction is reduced from 3 to 2, what will be the new Avg CPI? Also, calculate the new CPU time. **[3]**
d. Suppose the CPU's voltage and frequency were reduced by 10% and 20%, respectively, keeping the capacitance the same. How much will the power be reduced? **[2]**
e. Given the Wafer has a radius of 80mm and the Die area is 100 mm^2, calculate the number of dies per Wafer. **[2]**

a) $\text{Avg. CPI} = \dfrac{(1\times10^6\times2)+(2\times10^6\times3)+(4\times10^6\times1)}{7\times10^6}$

$= 1.71$

b) $\text{CPU Time} = \dfrac{\text{IC}\times\text{CPI}}{\text{Clock rate}} = \dfrac{7\times10^6\times1.71}{2.5\times10^9} = 4.79\text{ms}$

c) New Avg. CPI = $\frac{(1\times10^6\times2)+(2\times10^6\times2)+(4\times10^6\times1)}{7\times10^6}$

$= 1.43$

New CPU Time $= \frac{IC \times CPI}{\text{clock rate}} = \frac{7\times10^6 \times 1.43}{2.5\times10^9}$

$= 4\,ms$

d) $\frac{\text{Power New}}{\text{Power old}} = \frac{c \times v^2 \times f}{c \times v^2 \times f}$

$= \frac{C_{old} \times (0.9\, V_{old})^2 \times (0.8\, f_{old})}{C_{old} \times V_{old}^2 \times f_{old}}$

$\text{Power}_{new} = 0.648 \times \text{Power}_{old}$

Power reduced $= 1 - 0.648 = 0.352 = 35.2\%$

e) Dies per wafer $= \frac{\text{wafer Area}}{\text{Die Area}}$

$= \frac{\pi (80)^2}{100}$

$= \lfloor 201.06 \rfloor$

$= 201$ dies

BRAC UNIVERSITY
Inspiring Excellence

CSE340: Computer Architecture
Spring 2026
Quiz - 01
Duration: 25 minutes

B

Name: Solution	ID:	Section: 23

1. Write T (True) or F (False) beside the questions. 1 mark each. **[1 x 3 = 3]**

a. Increasing yield reduces the cost per die if wafer cost remains fixed. T

b. Two processors running at the same clock rate must have the same performance. F

c. "*Dependability via redundancy*" means using the least number of components for an efficient design F

2. An AMD CPU runs a Discord program with a clock rate of 2.5GHz that has the following instruction mix:

Instruction Class	Instruction Count	CPI
Load	$2x10^6$	3
Multiplication	$3x10^6$	1
Branch	$1x10^6$	2

f. Find the Average CPI for this program. **[3]**

g. Find the CPU execution Time. **[2]**

h. If the CPI of Load instruction is reduced from 3 to 2, what will be the new Avg CPI? Also, calculate the new CPU time. **[3]**

i. Suppose the CPU's voltage and capacitance were reduced by 10% and 20%, respectively, keeping the frequency the same. How much will the power be reduced? **[2]**

j. Given the Wafer has a radius of 80mm and the Die area is 100 mm^2, calculate the number of dies per Wafer. **[2]**

f) $$\text{Avg. CPI} = \frac{(2\times10^6\times3)+(3\times10^6\times1)+(1\times10^6\times2)}{(6\times10^6)}$$

$$= 1.83$$

g) $$\text{CPU Time} = \frac{IC \times CPI}{\text{clock rate}} = \frac{6\times10^6\times1.83}{2.5\times10^9}$$

$$= 0.004392\,s$$

$$= 4.39\,ms$$

h) New Avg. CPI $= \dfrac{(2\times10^6\times2) + (3\times10^6\times1) + (1\times10^6\times2)}{6\times10^6}$

$= 1.5$

New CPU Time $= \dfrac{IC \times CPI}{\text{Clock rate}}$

$= \dfrac{6\times10^6 \times 1.5}{2.5\times10^9}$

$= 3.6\text{ms}$

i) $\dfrac{P_{new}}{P_{old}} = \dfrac{C_{old}\times0.8 \times (V_{old}\times0.9)^2 \times f_{old}}{C_{old}\times(V_{old})^2\times f_{old}}$

$P_{new} = 0.648\ P_{old}$

Power reduced $= 1-0.648 = 0.352 = 35.2\%$

j) Dies per Wafer $= \dfrac{\text{Wafer Area}}{\text{Die Area}}$

$= \dfrac{\pi\times80^2}{100}$

$= \lfloor 201.06 \rfloor$

$= 201$ dies